Database Systems: Relational Model

The relational database model organizes data into one or more tables (or relations) of columns and rows, with a unique key identifying each row. Relational algebra provides the formal theoretical foundation for relational databases and SQL queries. A primary key uniquely identifies each record in a relational table, ensuring entity integrity.

Key Concepts:
- Concept Definitions
- Implementation Details
- Performance Considerations

Study Notes for Computer Science 301.

Transactions and ACID Properties

A database transaction symbolizes a unit of work performed within a database management system. Transactions must satisfy the ACID properties: Atomicity, Consistency, Isolation, and Durability. Atomicity guarantees that all transactions execute completely or not at all.

Key Concepts:
- Concept Definitions
- Implementation Details
- Performance Considerations

Study Notes for Computer Science 301.

Indexing and B-Trees

A database index is a data structure that improves the speed of data retrieval operations on a database table. B-Trees and B+ Trees are self-balancing search trees widely used in relational database storage engines for range queries and fast lookups.

Key Concepts:
- Concept Definitions
- Implementation Details
- Performance Considerations

Study Notes for Computer Science 301.

Concurrency Control: Two-Phase Locking

Two-Phase Locking (2PL) is a concurrency control method that guarantees serializability. In the growing phase, locks are acquired and no locks are released. In the shrinking phase, locks are released and no new locks can be acquired.

Key Concepts:
- Concept Definitions
- Implementation Details
- Performance Considerations

Study Notes for Computer Science 301.